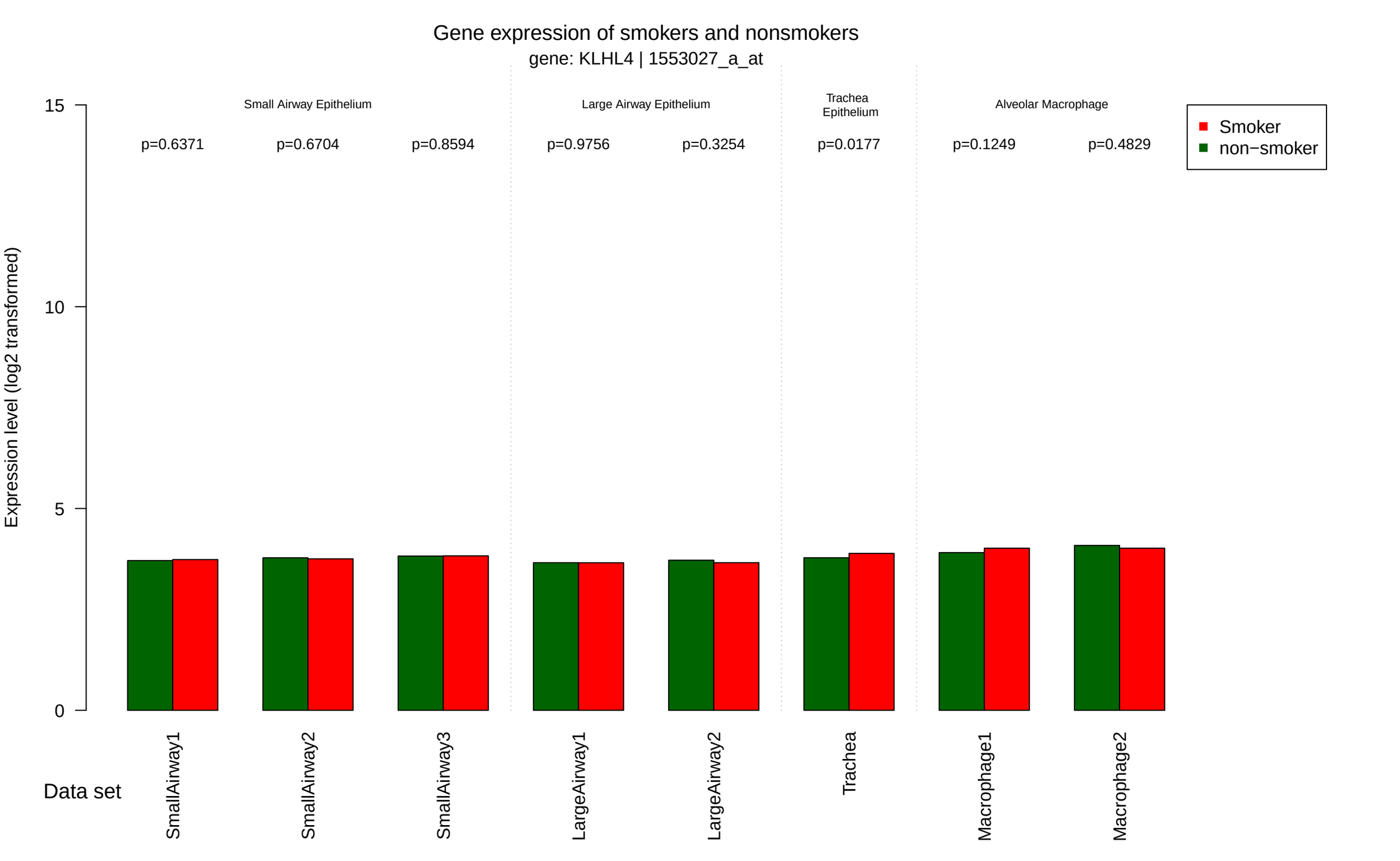
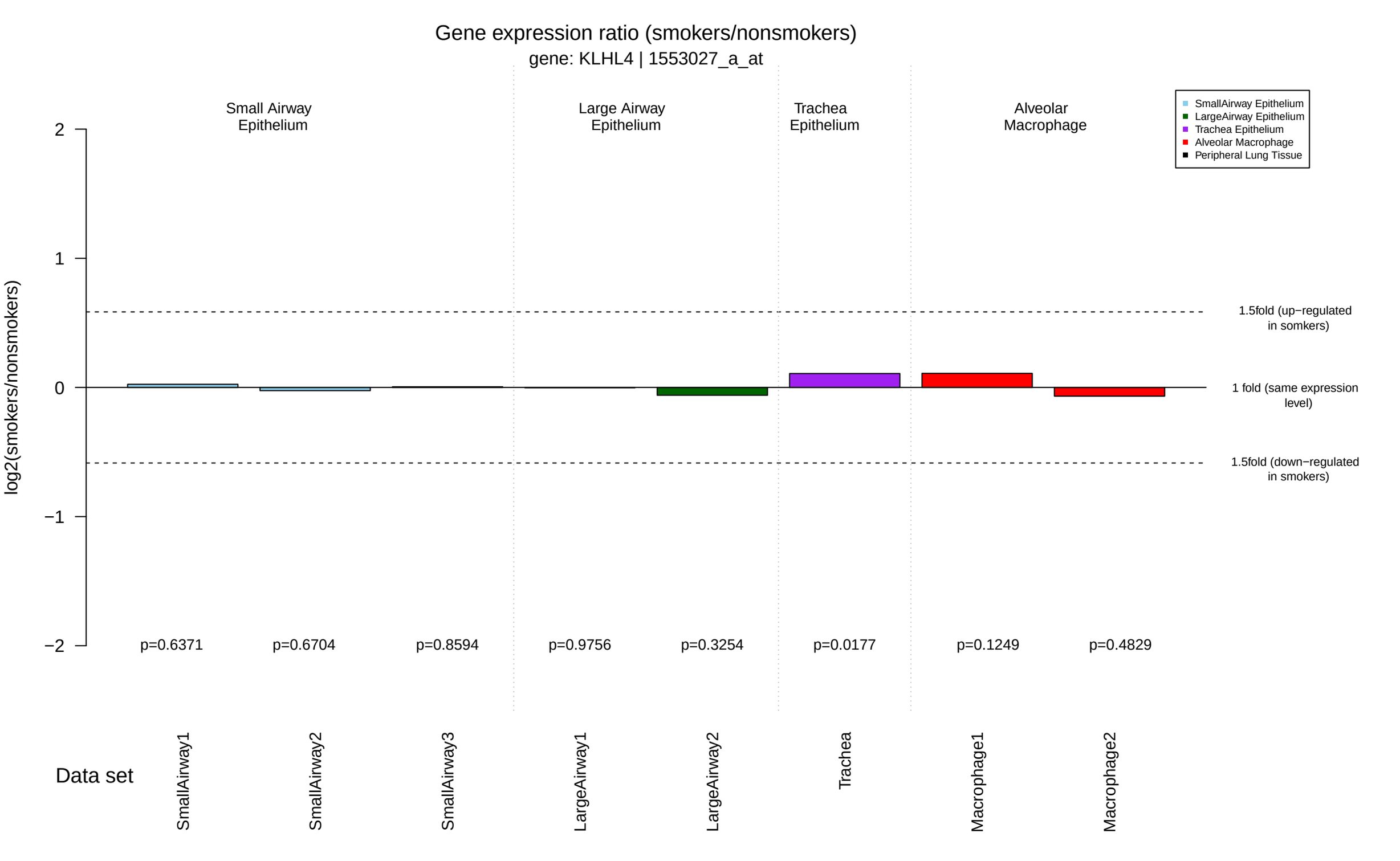
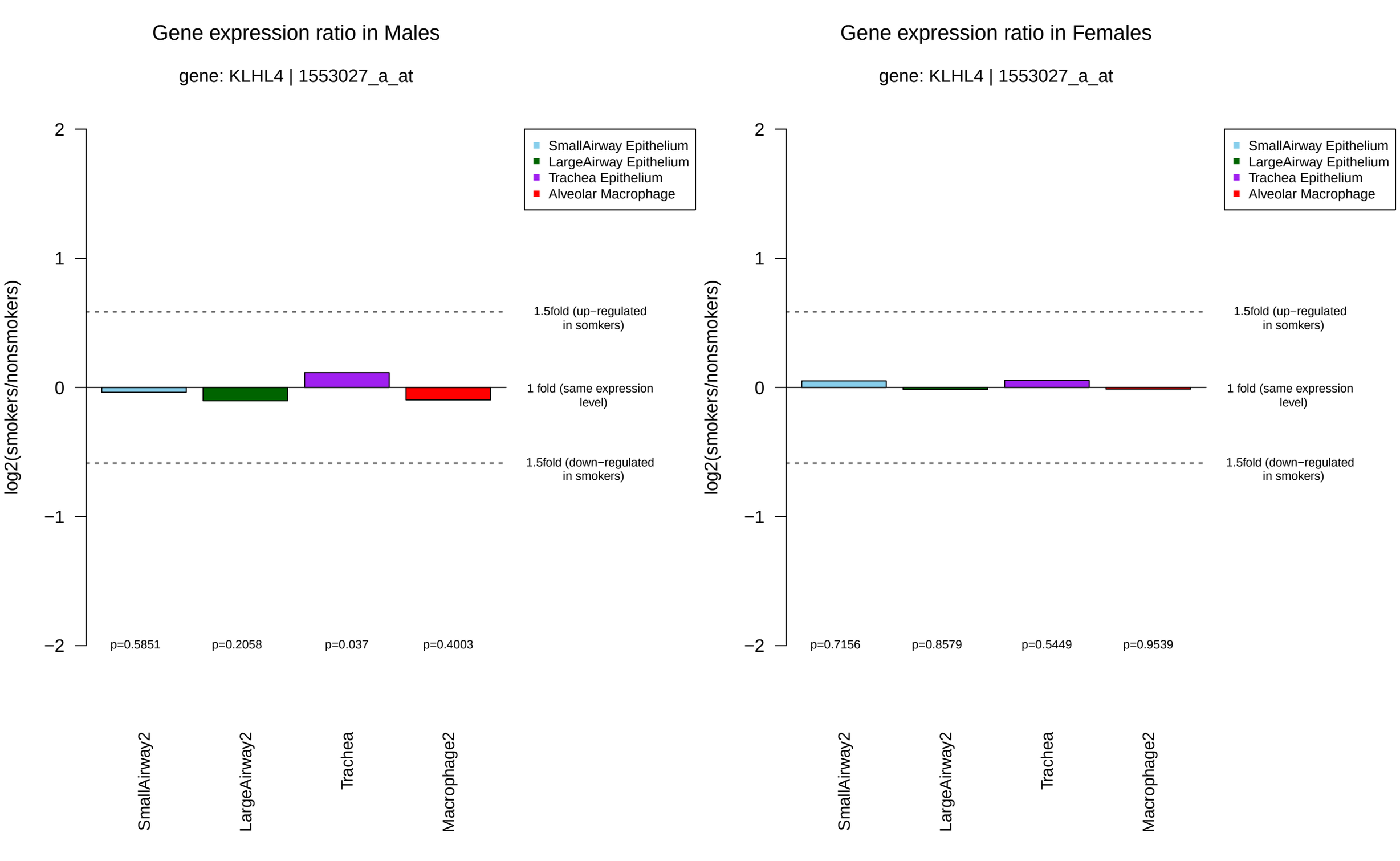
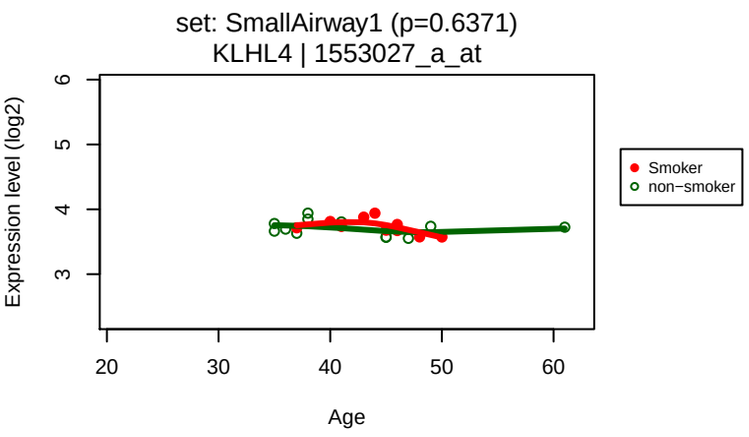
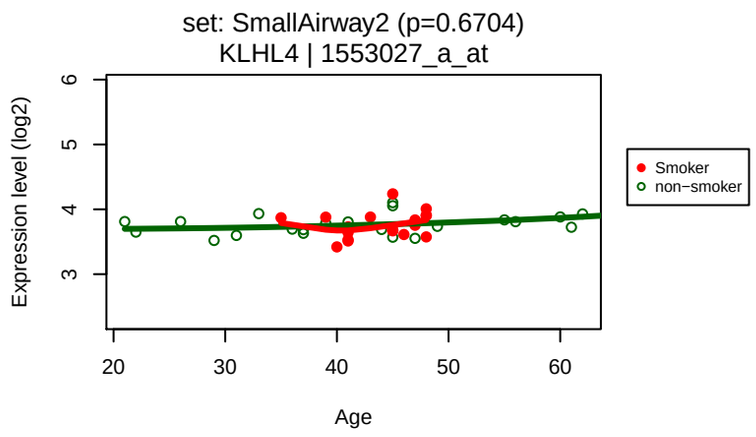
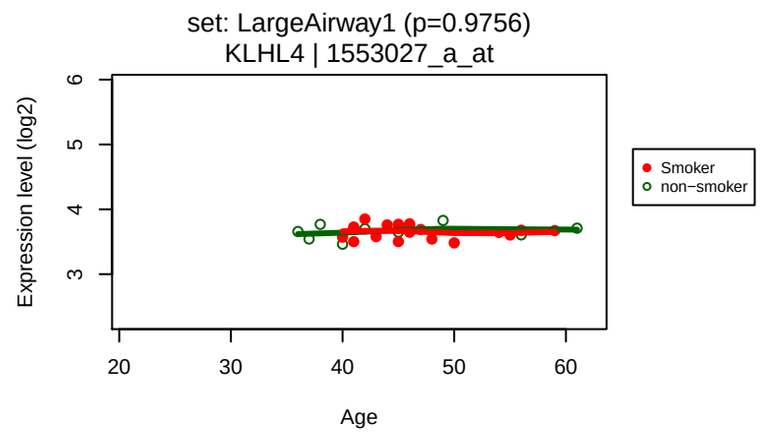

Gene expression of smokers and nonsmokers
gene: KLHL4 | 1553027_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6371
p=0.6704
p=0.8594
p=0.9756
p=0.3254
p=0.0177
p=0.1249
p=0.4829
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

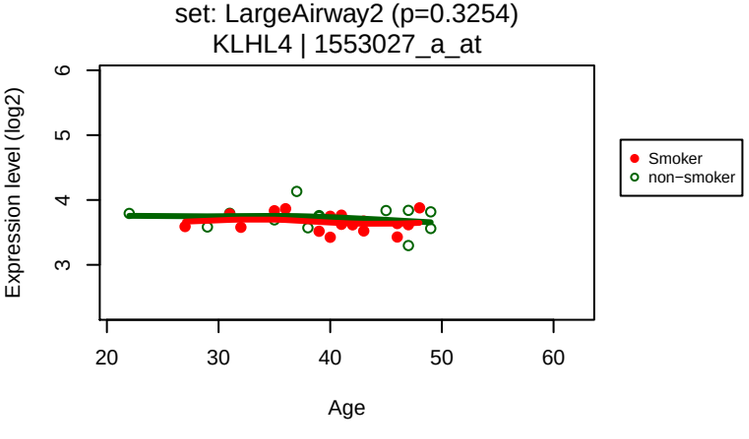

Gene expression ratio (smokers/nonsmokers)
gene: KLHL4 | 1553027_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6371
p=0.6704
p=0.8594
p=0.9756
p=0.3254
p=0.0177
p=0.1249
p=0.4829
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

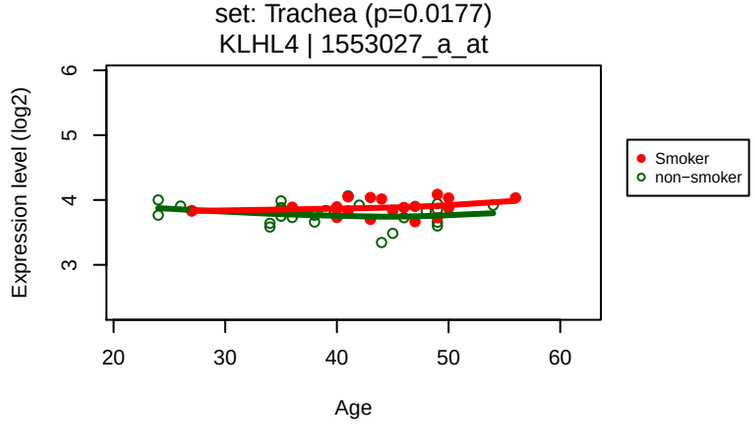

Gene expression ratio in Males
gene: KLHL4 | 1553027_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5851
p=0.2058
p=0.037
p=0.4003
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHL4 | 1553027_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7156
p=0.8579
p=0.5449
p=0.9539
SmallAirway2
LargeAirway2
Trachea
Macrophage2

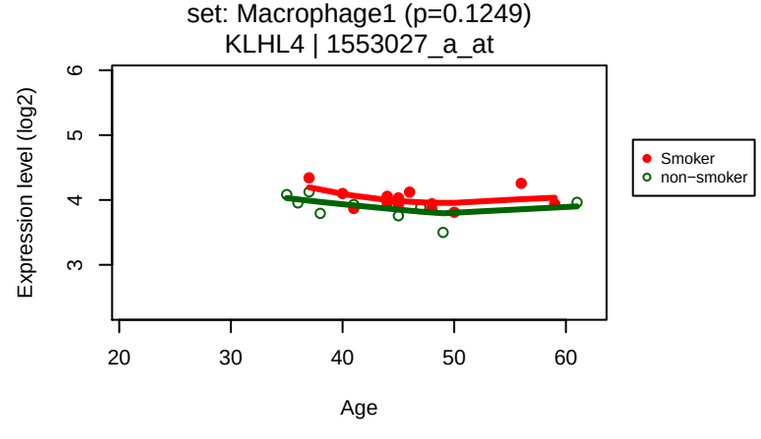

set: SmallAirway1 (p=0.6371)
KLHL4 | 1553027_a_at
Expression level (log2)
Age
Smoker
non−smoker

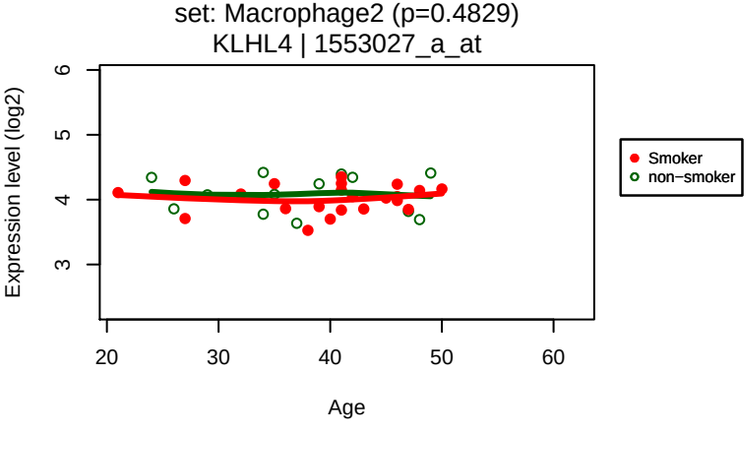

set: SmallAirway2 (p=0.6704)
KLHL4 | 1553027_a_at
Expression level (log2)
Age
Smoker
non−smoker

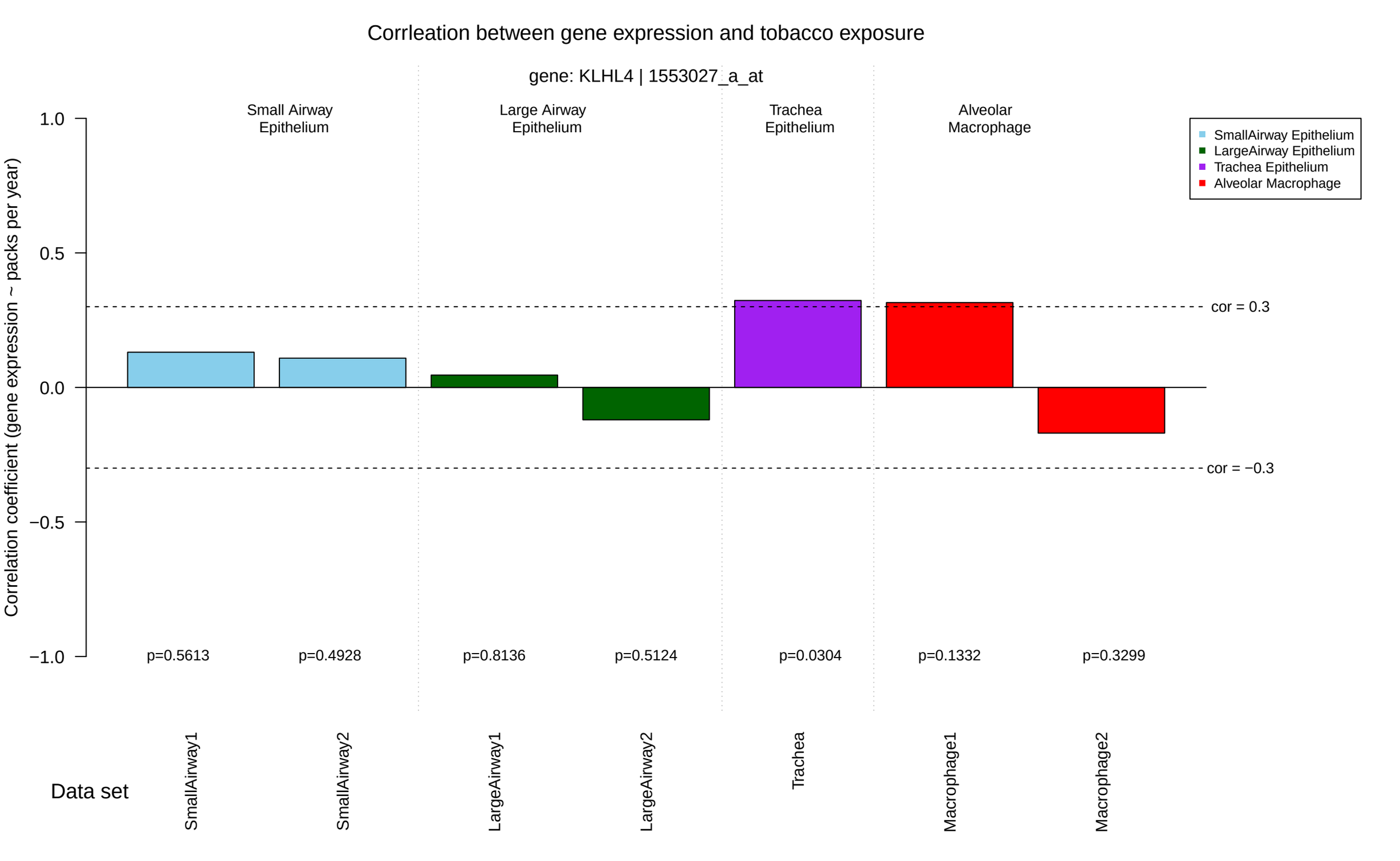

set: LargeAirway1 (p=0.9756)
KLHL4 | 1553027_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3254)
KLHL4 | 1553027_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0177)
KLHL4 | 1553027_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1249)
KLHL4 | 1553027_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4829)
KLHL4 | 1553027_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHL4 | 1553027_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5613
p=0.4928
p=0.8136
p=0.5124
p=0.0304
p=0.1332
p=0.3299
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2